

Bruce Shi

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SUMMARY OF QUALIFICATIONS

- Mechanical Engineering student focused on **CAD-driven product design, R&D prototyping, and manufacturing validation**, with experience across engineering design, quality/failure analysis, and full-cycle product development
- Applied **3D modeling, assembly design, GD&T, technical drawings, and BOM documentation** through industry experience with high-speed rail air supply units and personal design projects.
- Hands-on experience connecting CAD to fabrication through **3D printing, machining, and composite manufacturing**.

EXPERIENCE

Engineering Design Intern | KNORR-BREMSE

May 2025 – Aug 2025

- Supported product design and development of high-speed **rail air supply units** using Creo Parametric
- Created 500+ **CAD drawings** for parts and assemblies, includes dimensional drawings, exploded views, GD&T, BOM callouts
- Designed 100+ 3D models, including **sheet metal, piping, cabling, and welded components**, ensure fit and manufacturability
- Updated assemblies based on customer feedback and engineering change requests to support rapid design iterations
- Proficient with **PLM-based engineering workflows** (Windchill), including BOM structure, engineering document control, revision tracking, and engineering change management

Quality Assurance Intern | LINAMAR

Jan 2026 – Apr 2026

- Interpreted engineering drawings and GD&T to perform dimensional validation of automotive components using **CMM**
- **Designed inspection fixtures** in SolidWorks to improve measurement consistency, repeatability, and part validation workflow
- Conducted **root cause analysis**, linking defects to design limitations, material behavior, and manufacturing process variation
- Evaluated welded and heat-treated components through **metallurgical** inspection to identify defects and assess impacts
- Collaborated with cross-functional teams to improve production quality, contributing to a **5% scrap reduction**

Baja SAE Design Team – Chassis Subteam | UNIVERSITY OF WATERLOO

Sept 2025 – Present

- Designing the vehicle chassis in SolidWorks following SAE Baja rules, focusing on strength, weight, and manufacturability.
- Assisted in cutting and machining frame components in the machine shop to support prototype fabrication.

SELECTED PROJECTS

Custom Ergonomic Carbon Fiber Mouse | *Personal Project* | *Full-cycle Prototyping*

Jan 2026 – Apr 2026

- Designed a Ergonomic mouse, converting hand geometry into a manufacturable CAD model using Blender & SolidWorks
- Reverse-engineered electronics from a Logitech M546 mouse and designed internal mounts with **0.4 mm fit accuracy**
- Iterated **10+ FDM 3D-printed prototypes** to validate ergonomics, component fit, and assembly feasibility
- Designed molds and fabricated a carbon fiber composite shell, **reducing weight by 20%** while maintaining strength

Carbon Fiber Case | *Personal Project* | *Mold Design & Composite Manufacturing*

June 2025 - Aug 2025

- Designed multi-piece molds for the AirPods case in Creo, applying GD&T to ensure proper fit for 4 carbon-fiber layers.
- Used carbon fiber sheets with AB epoxy resin, applied **hand layup** and **vacuum bagging** techniques.

V12 Engine CAD & Simulation | *Personal Project*

May 2025 - June 2025

- Modeled **50+ components** of a V12 engine in SolidWorks from scratch and assembled them with correct mates.
- Conducted a kinematic simulation using Motion Study to replicate the piston-crankshaft motion and valve operation.

McLaren 720S Spider – High-Fidelity 3D Modeling | *Personal Project*

Jan 2024 - Jan 2025

- Modeled 100 + components in Blender, built realistic geometry referencing blueprints to strengthen surface modeling and visualization skills for CAD. Currently developing Cycles rendering workflow.

SKILLS

CAD & Simulation: SolidWorks, Creo, AutoCAD, CATIA, Ansys, Blender

Design & Validation: GD&T, DFM, tolerance/fit validation, BOM management, root cause analysis, failure analysis

Fabrication & Testing: FDM/SLA 3D printing, CNC machining, laser cutting, CMM, composite manufacturing

Technical Knowledge: heat treatment effects, weld defects, tolerance stack-up, ISO standards, PLM workflow

EDUCATION

University of Waterloo | *BASc in Honours Mechanical Engineering*

Sept 2024 - April 2029

- Cumulative GPA: 89.94%
- Relevant course: Control of Material Properties, Electromechanical Devices & Power Processing